

4.2 QB - loops in java [for loop Advanced]

♦ Basic Number Operations

1. Print numbers from 1 to 10 using a `for` loop.
 2. Print numbers from 10 to 1 in reverse using a `for` loop.
 3. Print even numbers from 1 to 50.
 4. Print odd numbers from 1 to 50.
 5. Print squares of numbers from 1 to `n` (user input).
 6. Print cubes of numbers from 1 to `n`.
 7. Print all numbers divisible by 3 between 1 and 30.
 8. Print numbers from 1 to `n` but skip multiples of 5.
 9. Print numbers in reverse from `n` to 1 (user input).
 10. Print multiplication table for a number using `do-while`.
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♦ Sum, Count & Math Operations

11. Find the sum of numbers from 1 to `n`.
12. Find the sum of even numbers between 1 and `n`.
13. Find the sum of odd numbers between 1 and `n`.
14. Count the number of digits in a number.
15. Calculate the sum of digits of a number.
16. Calculate the factorial of a number using `for`.
17. Reverse the digits of a number (e.g., 456 → 654).
18. Find the product of digits in a number.
19. Find the maximum digit in a number.
20. Find the minimum digit in a number.

♦ User Interaction & Input Validation

21. Keep asking the user to enter a positive number until they do so.
22. Ask the user to enter a number greater than 100; repeat if input is smaller.
23. Keep asking for a username until "admin" is entered.
24. Repeat a menu until the user chooses "Exit".
25. Keep taking marks input until the user enters -1, then calculate average.
26. Ask for a password; allow a maximum of 3 tries.

27. Keep asking for a number until a prime number is entered.
 28. Ask the user for a valid age between 1–120.
 29. Let the user input numbers until they enter an even number.
 30. Keep generating random numbers until one is divisible by 7.
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♦ **Number-Based Logic (Non-repetitive from earlier)**

31. Find the first number greater than 100 that is divisible by both 4 and 6.
 32. Print the first 10 numbers divisible by 9.
 33. Count how many digits in a number are odd.
 34. Sum all digits of a number until a single digit is obtained (repeated sum).
 35. Reverse a number and check if the reversed number is even.
 36. Print digits of a number from right to left using `for`.
 37. Print digits of a number from left to right using `for`.
 38. Convert a binary number (as int) to decimal using a loop.
 39. Convert a decimal number to octal using a loop.
 40. Check if a number has more even digits or odd digits.
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♦ **Series and Patterns**

41. Print the series: 2 4 8 16 32 ... up to `n` terms.
 42. Print the sum of the series: $1 + 1/2 + 1/3 + \dots + 1/n$
 43. Generate and print the first `n` terms of the Tribonacci series.
 44. Create a star pattern of fixed rows using a `for` loop.
 45. Create a triangle number pattern using `for`.
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♦ **Fun / Game-Like Logic**

46. Simulate a simple number guessing game using `for`.
 47. Flip a virtual coin until you get "Heads" 3 times.
 48. Generate random numbers until the total sum exceeds 100.
 49. Create a number lock: keep asking until the correct 4-digit PIN is entered.
 50. Simulate a dice roll until you roll a 6.
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